SUBSEA MATTRESS VERIFICATION REPORT: CR-A1_M16 (#A007)

Project: Tung-Hsiao Power Plant 2nd Stage Renewal Subsea Gas Pipeline Project
Client: Taiwan Power Company (TPC) | Contractor: NMDC Energy | Vessel: DP2 MMA Prestige
Crossing Array: CR-A1_L1 | Location ID: CR-A1_M16 | Mattress ID: #A007 | Active USBL: Survey Pole
Survey Date/Time: 2026-11-05 04:20 | Deliverable Base: 20261105_Mattress_CR-A1_M16_A007
Chart Scales @ A3: Panel 1 (1:250) | Panel 2 (1:75) | Panel 3 (1:75/1:15) | Panel 4 (1:50/1:15)

EXECUTIVE VERDICT

- Overall Status: VERIFIED - WITHIN DESIGN & GAP TOLERANCE
- Identification & Sensor: Mattress #A007 at CR-A1_M16 (CR-A1_L1) tracked via Survey Pole.
- Centerline Offset (dCL): 0.180 m vs. Tolerance +/-0.50 m (PASS) [dAlong: +0.179 m, dCross: -0.023 m].
- Heading Alignment: Design South Brg 195.854 deg vs. As-Laid 198.393 deg (dBrg: +2.539 deg).
- Surveyor Corner Gaps: NW:0.02m | SW:0.10m | Max Gap 0.10 m vs. Side-to-Side Gap Tolerance <=0.50 m (PASS).

1. GEODETIC, 4-CORNER ROV & GAP COMPARISON TABLE

Parameter	Design Baseline	QA/QC USBL Fix	As-Laid 4-Corner ROV	Delta / Status
Easting E (m, TWD97)	194544.329	194544.356	194544.356	+0.027 m
Northing N (m, TWD97)	2702612.575	2702612.753	2702612.753	+0.178 m
Depth Z (m, TWVD2001)	Seabed + 0.30 m	-	-18.35	Verified WROV
South Bearing (deg)	195.854 deg	198.397 deg	198.393 deg	+2.539 deg
North Azimuth (deg)	15.854 deg	-	18.393 deg	+2.539 deg
Radial Offset dCL (m)	0.000 m (Tol +/-0.50m)	-	0.180 m	PASS (91.0% in box)
Side-to-Side Gap (m)	Tol <=0.50 m	-	NW:0.02m	SW:0.10m	PASS

2. KEY SURVEY FINDINGS (4-CORNER ROV COORDINATES & CORNER GAPS)

- C1 (NW Corner): E 194543.879 m, N 2702616.073 m, Z -18.35 m | Surveyor Corner Gap NW: 0.02
- C2 (NE Corner): E 194546.726 m, N 2702615.126 m, Z -18.35 m | Surveyor Corner Gap NE: N/A
- C3 (SE Corner): E 194544.833 m, N 2702609.433 m, Z -18.35 m | Surveyor Corner Gap SE: N/A
- C4 (SW Corner): E 194541.986 m, N 2702610.380 m, Z -18.35 m | Surveyor Corner Gap SW: 0.10
- Surveyor Recorded Corner Gaps (Tol <=0.50m Side-to-Side Gap): NW:0.02m | SW:0.10m.
- Environment: Wind: Recorded in Log | Current: Recorded in Log.

3. SURVEYOR REMARKS

- [Survey Pole] Mattress installed completed (CR-A1_M16) #A007 (04:20 - 04:40); Gap between Mat is 2cm North a
  nd 10cm South; ROV 4-corner fix completed at 05:00

4. HISTORICAL PLACED UNITS LEDGER

Unit #	Date/Time	Location ID	Tag ID	As-Laid E	As-Laid N	Brg (deg)	dCL (m)	Corner Gaps
1	2026-11-04 14:34	CR-A1_M15	#A046	194541.206	2702613.513	196.89	0.265	1st Unit (N/A)